

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: When you scale up a shape or solid — like doubling the size of a water tank or enlarging a building plan — what exactly happens to its lengths, its area, and its volume, and how can we use this to solve real problems in our community?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Properties of Similar Figures	Students examined pairs of figures (e.g., a small and large version of a rectangular maize store floor plan from a Rift Valley farm) and measured corresponding angles and side lengths. They recorded angle measures and computed side ratios in a table, noticing whether the ratios were consistent across all pairs of corresponding sides.	Two figures are similar if and only if (1) all corresponding angles are equal AND (2) all corresponding sides share the same ratio — the Linear Scale Factor ($LSF = \text{image length} \div \text{object length}$). Students learned to identify corresponding vertices, write similarity statements with correct notation (e.g., $\triangle ABC \sim \triangle DEF$), and calculate the LSF from any pair of corresponding sides.	PEDAGOGICAL NOTE — This is the anchor lesson: keep returning to the maize-store scenario throughout the unit. Key answer-key points: (a) Similarity requires BOTH equal angles AND proportional sides — one condition alone is insufficient. (b) LSF is always expressed as image:object; $LSF > 1$ means enlargement, $0 < LSF < 1$ means reduction. (c) Common student error: dividing object by image instead of image by object — address this explicitly. Prompt: 'If the small store is 4 m × 6 m and the large store is 6 m × 9 m, what is the LSF?' Expected answer: $9/6 = 6/4 = 1.5$. Use cold-call after 30 s wait time on at least 3 students before confirming.
Lesson 2 Identifying and Using the Linear Scale Factor (LSF)	Students were given sets of similar shape pairs drawn on plain paper — including a small and an enlarged version of a Kenya outline map and a Kenyan school building blueprint — and tasked with measuring corresponding lengths using rulers. They recorded all measurements and computed ratios for every pair of corresponding sides to verify consistency of the LSF.	The LSF is a single constant ratio that relates every pair of corresponding lengths in two similar figures. Students practised: (1) calculating the LSF from given measurements; (2) using LSF to find unknown side lengths (unknown = known corresponding side × LSF); (3) distinguishing between $LSF > 1$ (enlargement), $LSF = 1$ (congruence), and	PEDAGOGICAL NOTE — Emphasise the multiplicative (not additive) nature of similarity. A common misconception is that sides grow by a fixed additive amount rather than a fixed multiplier. Key worked example: Object sides 3 cm, 5 cm, 7 cm; $LSF = 2.5 \rightarrow$ Image sides 7.5 cm, 12.5 cm, 17.5 cm. Perimeter of object = 15 cm; perimeter of image = 37.5 cm; ratio = 2.5 =

		LSF < 1 (reduction). They also confirmed that the LSF applies equally to perimeters: Perimeter ratio = LSF.	LSF ✓. Tie back to the driving question: 'If a building plan uses a scale of 1:50, what is the LSF when going from plan to real building?' (LSF = 50.) Wait 45 s, cold-call 3 students. Ensure students can also work backwards: given an image side and the LSF, find the object side.
Lesson 3 Constructing Images under Enlargement on a Plane Surface	Students used a ruler and a fixed centre of enlargement (point O marked on plain paper) to construct the image of a triangle by drawing rays from O through each vertex and marking the image vertices at distances LSF × (object distance from O). They performed constructions for LSF = 2 and LSF = 3, each time measuring to verify image sides were exactly k times object sides.	When a polygon is enlarged with centre O and LSF k: (1) every image vertex lies on the ray from O through the corresponding object vertex, at distance $k \times OA$ from O; (2) every image side = $k \times$ corresponding object side; (3) a positive k places the image on the SAME side of O as the object; (4) $k > 1$ produces a larger image farther from O; $0 < k < 1$ produces a smaller image closer to O. Students learned the construction algorithm: draw ray → measure OA → mark $OA' = k \times OA$ → repeat for all vertices → join image vertices.	PEDAGOGICAL NOTE — Stress that the centre of enlargement is the invariant point: it does NOT move. A useful check students should always apply: extend lines through corresponding object and image vertices — they must all pass through O. Common construction error: measuring $k \times$ (side length) rather than $k \times$ (distance from O to vertex). Worked example answer-key: Object $\triangle ABC$ with $OA = 3$ cm, $OB = 4$ cm, $OC = 5$ cm; $k = 2 \rightarrow OA' = 6$ cm, $OB' = 8$ cm, $OC' = 10$ cm. Contextualise: 'A jua kali artisan in Kamukunji, Nairobi, is scaling up a metal gate design by factor 3 — how far must each corner move from the reference point?' Use this as the exit problem; wait 1 minute before cold-calling.
Lesson 4 Enlargement on the Cartesian Plane: Positive and Negative Linear Scale Factors	Students plotted object polygons on squared Cartesian paper and applied the enlargement formula Image Point = Centre + $k \times$ (Object Point – Centre) for various centres and scale factors, including negative values ($k = -2$). They compared the positions and orientations of images produced by positive versus negative LSFs.	On the Cartesian plane: (1) Enlargement formula: $P' = C + k(P - C)$, where C is the centre and P is any object point. (2) When the centre is the origin, this simplifies to $P' = kP$, i.e., $(x, y) \rightarrow (kx, ky)$. (3) A POSITIVE k produces an image on the SAME side of the centre as the object (same orientation). (4) A NEGATIVE k produces an image on the OPPOSITE side of the centre — the image is also rotated 180° (point reflection through C). (5) $ k $ determines the size change; the sign determines the side/orientation.	PEDAGOGICAL NOTE — The negative LSF is the most common source of error in KCSE questions. Drill the formula with a clear worked example: Centre C = (1, 2), $k = -2$, Object point P = (3, 4). $P' = (1, 2) + (-2)[(3, 4) - (1, 2)] = (1, 2) + (-2)(2, 2) = (1, 2) + (-4, -4) = (-3, -2)$. Emphasise: $ k = 2$ so image is twice as large; negative sign so image is on the opposite side of C. Provide a 'positive vs negative k' side-by-side diagram on the board for student reference. Contextual hook: 'A surveyor in Kisumu uses a negative scale factor on a map grid to flip and enlarge a plot boundary — what happens to the plot's position?' Cold-call 4 students after 45 s wait.
Lesson 5 Area Scale Factor: Discovering How Area Grows When You Enlarge a Shape	Students drew a 2 cm × 3 cm rectangle (object) on squared paper, then drew enlargements with LSF = 2, 3, and 4. They counted grid squares to find the area of each image, recorded the results in a table	The Area Scale Factor (ASF) equals the SQUARE of the Linear Scale Factor: $ASF = (LSF)^2 = k^2$. Doubling all lengths QUADRUPLES area ($2^2 = 4$); tripling all lengths multiplies area by 9 ($3^2 = 9$). This	PEDAGOGICAL NOTE — Use a unit-square argument to build intuition: a 1×1 square has area 1; scaled by k it becomes $k \times k$, area k^2 — this is why the exponent is 2. Key worked example (answer key): A

	(LSF, Object Area, Image Area, Image/Object Area ratio), and looked for a pattern in the ratio column.	holds for ALL similar figures — triangles, circles, irregular shapes — not just rectangles. Students learned to: (1) find ASF from LSF: $ASF = k^2$; (2) find LSF from ASF: $LSF = \sqrt{ASF}$; (3) calculate an unknown area given the other area and the LSF.	triangular vegetable garden in Kericho has area 48 m^2 ; a similar garden has $LSF = 1.5$ relative to it. Image area = $48 \times (1.5)^2 = 48 \times 2.25 = 108 \text{ m}^2$. Reverse problem: two similar plots have areas 75 m^2 and 300 m^2 — find LSF. $ASF = 300/75 = 4 \rightarrow LSF = \sqrt{4} = 2$. Common error: students multiply area by k instead of k^2 — anticipate and address with a direct counter-example. Connect to driving question: 'If we double a water tank's base dimensions, what happens to the base area?'
Lesson 6 Volume Scale Factor and the LSF–ASF–VSF Relationship	Students built unit-cube models using interlocking cubes (or sketched them on isometric paper) representing a $1 \times 1 \times 1$ cube, then a $2 \times 2 \times 2$ cube, then a $3 \times 3 \times 3$ cube. They counted unit cubes to find volume and recorded LSF, Surface Area (counting faces), and Volume in a table, computing ASF and VSF ratios.	The Volume Scale Factor (VSF) equals the CUBE of the Linear Scale Factor: $VSF = (LSF)^3 = k^3$. The complete three-way relationship is: $LSF = k \mid ASF = k^2 \mid VSF = k^3$. Doubling linear dimensions multiplies volume by $8 \text{ (} 2^3 \text{)}$; tripling multiplies volume by $27 \text{ (} 3^3 \text{)}$. To find LSF from VSF: $LSF = \sqrt[3]{VSF}$. Students practised all algebraic directions: given any one of {LSF, ASF, VSF}, find the other two.	PEDAGOGICAL NOTE — Use the water-tank anchor explicitly here: 'A cylindrical water tank in Machakos village has height 1 m and radius 0.5 m. A second, similar tank is scaled up by $LSF = 2$. The new tank holds $2^3 = 8$ times as much water.' This is the emotional hook of the unit. Key answer-key table: $k=2 \rightarrow ASF=4, VSF=8$; $k=3 \rightarrow ASF=9, VSF=27$; $k=1/2 \rightarrow ASF=1/4, VSF=1/8$. Common exam trap: ASF and VSF problems that give mass or capacity instead of volume — remind students that $\text{mass} \propto \text{volume}$ for uniform density, so mass also scales by k^3 . After 1-minute pair-discussion, cold-call at least 4 students to state the VSF for given LSF values before confirming answers.
Lesson 7 Application to Real-Life Situations & Model-Building Project Launch	Student groups were given real-life scenario cards set in Kenyan contexts (e.g., a Kisumu fish-drying shed being scaled up, a Nairobi apartment balcony tile pattern, two similar water tanks with a given capacity ratio) and worked through multi-step problems requiring them to determine LSF from given areas or volumes, then find missing lengths, areas, or volumes. Groups also received their model-building project brief.	Students confirmed the integrated application of all three scale factors: when $LSF = k$, area scales by k^2 and volume scales by k^3 . They practised the full problem-solving pathway: (1) identify whether given information yields LSF, ASF, or VSF directly; (2) convert to LSF if necessary ($LSF = \sqrt{ASF}$ or $LSF = \sqrt[3]{VSF}$); (3) use LSF to find missing lengths; (4) use k^2 or k^3 to find missing areas or volumes. The taller water tank with $k = 2$ holds $2^3 = 8$ times the volume — not 2 times, not 4 times.	PEDAGOGICAL NOTE — This is the synthesis lesson; insist students show their $LSF \rightarrow ASF \rightarrow VSF$ reasoning chain explicitly, not just final answers. Worked answer-key example: Two similar conical grain stores near Eldoret — the smaller has height 1.5 m, base area 7 m^2 , volume 3.5 m^3 . The larger has height 3 m. Step 1: $LSF = 3/1.5 = 2$. Step 2: $ASF = 4 \rightarrow$ larger base area = 28 m^2 . Step 3: $VSF = 8 \rightarrow$ larger volume = 28 m^3 . Common error: students use the LSF to scale volume directly ($\times 2$ instead of $\times 8$). Circulate during group work; if a group uses the wrong scale factor, ask 'What type of measurement is volume — 1D, 2D, or 3D?' to guide self-correction without giving the answer. Project brief should be distributed

			and explained in the final 10 minutes.
Lesson 8 Project Presentations, Model Exhibition & Unit Consolidation — Final Explanation	Groups presented their scale models and/or posters (e.g., scaled water tanks, building plans, grain-store comparisons) to the class, explaining their chosen LSF, the resulting ASF and VSF, and a real-world implication (cost, capacity, material use). Peers asked questions using a structured gallery-walk protocol. The class then completed a unit consolidation task linking back to the original anchoring phenomenon.	When any solid is enlarged by a Linear Scale Factor k: the Area Scale Factor is k^2 (surface area, cross-sectional area, base area all scale by k^2) and the Volume Scale Factor is k^3 (capacity, mass for uniform density, and volume all scale by k^3). These relationships explain why a water tank that is twice as tall and wide holds EIGHT times as much water, why a building that is three times the linear size of a model requires NINE times the roofing material, and why a grain store scaled up by factor 2.5 needs $2.5^3 = 15.625$ times the storage volume. The LSF–ASF–VSF chain ($k : k^2 : k^3$) is the master tool for all similarity and enlargement problems.	PEDAGOGICAL NOTE — This lesson serves as the formal answer key for the entire unit's driving question. During presentations, use a class evaluation rubric and listen for three key mathematical statements: (1) explicit statement of LSF with units or ratio; (2) correct derivation of $ASF = k^2$ with a numerical check; (3) correct derivation of $VSF = k^3$ with a numerical check and a real-world interpretation. If a group omits any of these, ask a targeted question: 'You told us lengths doubled — so what happened to the surface area of your tank?' After all presentations, conduct a whole-class 'DQB Final Review': revisit each question posted on the Driving Question Board over the unit and confirm it is now fully answered. Close with the summary statement: 'Scale lengths by k, areas by k^2, volumes by k^3 — and you can solve almost any scaling problem in engineering, agriculture, or architecture in Kenya or anywhere.'

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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